ModelIt!

Teacher's Guide

What Do Plants Need to Grow?

Water, Light, and Happy Plants

Quick Reference

Grade Level	Kindergarten
NGSS Standard	K-LS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: K-LS1-1

Use observations to describe patterns of what plants and animals (including humans) need to survive.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students set up a simple plant growth investigation comparing different amounts of water and observing the results over time.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms** — All animals need food in order to live and grow. They obtain their food from plants or from other animals. Plants need water and light to live and grow.
Crosscutting Concept	**Cause and Effect** — Students observe that changing the amount of water (cause) changes how well the plant grows (effect).

CAST Testing Alignment

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Plants are living organisms that need certain things from their environment to survive and grow. Understanding plant needs is foundational to biology and connects to food systems, ecology, and environmental stewardship.

What Do Plants Need?

Plants need four basic things to survive: water, sunlight, air, and nutrients from soil. Water is absorbed through roots and transported up through the stem to every part of the plant. Sunlight provides the energy plants use for photosynthesis — the process of making their own food from light, water, and carbon dioxide. Without adequate water or sunlight, plants cannot make food and will stop growing, wilt, and eventually die.

How Water Helps Plants Grow

Water plays several critical roles in plant growth. It carries dissolved minerals and nutrients from soil into the roots and throughout the plant. Water pressure inside plant cells keeps stems and leaves firm and upright — when a plant lacks water, cells lose pressure and the plant wilts. Water is also a key ingredient in photosynthesis. Most kindergarten-appropriate plants (grass seeds, bean sprouts) show visible growth differences within 5-7 days when given different amounts of water.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify Amount of Water as the external component (we decide how much to water) and Plant Growth as the internal component (the plant grows on its own based on how much water it gets).

E **Establish Relationships**
Students discover that more water leads to more Plant Growth — a positive relationship (up to a point).

V **Visualize & Model**
Students build a simple two-part model connecting Amount of Water to Plant Growth with a positive arrow.

E **Evaluate Outcomes**
Students test Little Water vs. Lots of Water over several days and observe growth differences.

R **Revise & Extend**
Students explore what happens when they change sunlight instead of water — do plants in the dark grow well even with water?

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Have you ever planted a seed? What did you do to help it grow? Today we will learn what plants really need!"*

Watch For

Show a live plant and a wilted plant (or pictures). Ask: Which plant looks happy? Which looks sad? Why?

Slide 2: What Will We Learn?

3 min

Say: *"Today we will discover what plants need to grow big and strong! Hint: one thing comes from the sky and one comes from the faucet!"*

Watch For

Introduce 'nutrients' and 'sunlight.' Have students act like a plant — crouch down small, then grow tall with arms out like leaves reaching for sun.

Slide 3: Let's Be Scientists!

3 min

Say: *"Our big question is: What does a plant need to grow big and strong? Can a plant grow with NO water? Let us find out!"*

Watch For

Students share with a partner what they think plants need. Make a class list on the board.

Slide 4: What Are the Parts?

5 min

Say: *"Our model has two parts. Amount of Water is how much water we give the plant — WE decide that! Plant Growth is how big the plant gets — the plant does that on its own."*

Watch For

Use sorting mats. Students place Amount of Water on OUTSIDE and Plant Growth on INSIDE. Pretend to water a plant with a cup.

Slide 5: How Are They Connected?

5 min

Say: *"When a plant gets MORE water, it grows BIGGER. When it gets LESS water, it grows smaller or wilts. That is a POSITIVE connection!"*

Watch For

Show pictures of a well-watered plant vs. a dry plant. Students give thumbs up for positive connection. Water a class plant together.

Slide 6: What Did We Find Out?

5 min

Say: *"Plants NEED water to grow! Water goes in through the roots and helps the plant make food and stand up tall. No water means a sad, droopy plant."*

Watch For

Chant: 'When Water goes up, Plant Growth goes up!' Complete the sentence frame. Students draw a well-watered plant.

Slide 7: STEM Challenge

10 min

Say: *"Now YOU are the gardeners! Plant your seeds and make a watering plan. How much water will you give your plant each day to help it grow the tallest?"*

Watch For

Distribute cups with soil, seeds (fast-growing grass or bean), water cups with marked amounts. Students plant, water, and draw predictions of how tall their plant will grow.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Amount of Water
- **INTERNAL:** Plant Growth

Relationships

- **Amount of Water to Plant Growth:** POSITIVE (+) — When a plant gets more water, it grows bigger and healthier because water carries nutrients and helps the plant make food through photosynthesis.

Key Discoveries

- Plants that get enough water grow taller and look healthier
- Plants that get very little water grow slowly and can wilt or turn brown
- Water helps carry nutrients from the soil up into the plant so it can grow

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants eat dirt"	Plants do not eat soil. They absorb water and tiny nutrients dissolved in the water through their roots. Plants actually make their own food from sunlight, water, and air through a process called photosynthesis.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"More water is always better"	While plants need water, too much water can actually hurt them. Over-watering can drown the roots because they also need air. The right amount of water — not too much and not too little — is best for plant growth.